

Ergodic Control for Robotic Exploration with an Adaptive Gaussian Mixture Model

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Abstract—The problem of exploring unknown spaces with robots arises in many real-world applications. While different approaches exist, those utilizing sensory data, e.g., vision, positioning, etc., have emerged as of particular interest. Among these, ergodic control is significant, as it accounts for both motion cost and optimality. However, ergodic controllers alone often encounter limitations when applied to unknown areas, as they require a predefined information measure. This paper introduces a general and model agnostic ergodic control methodology designed to overcome these limitations. Unlike traditional ergodic controllers, our approach dynamically derives an information measure from sensory data and derives a degree of obstacle avoidance as a “by-product.” Empirical data from simulations in diverse environments, including those featuring dense vegetation, precision agriculture, etc., demonstrate the effectiveness of our model agnostic ergodic controller, showcasing the applicability of our methodology compared to conventional techniques typically restricted to spaces where an information measure is readily available.

I. INTRODUCTION

Exploring unknown spaces with robots is a problem commonly addressed by different methodologies arising in computing and robotics. This is a recurring problem in many real-world use cases such as monitoring, reconstruction, exploration, etc., where robots are expected to cover a given space while performing an assigned task. A key challenge to the practical applicability of these methodologies is that of leveraging sensing and computing resources while, at the same time, maximizing the information gathered and optimizing the exploration accordingly [1, 2]. While there are different approaches in the literature, approaches that are informed by sensory data have emerged as of particular interest. Among these approaches, ergodic control is a significant result, as it provides a more natural way of searching through deterministic exploratory behaviors while accounting for both the motion cost and optimality [3].

Ergodic control is a planning and controls methodology that derives robot trajectories maximizing a given information measure, i.e., a distribution that encodes where information is expected to be found, so that robots spend more time in areas with high information while quickly traversing areas with low information [4–6]. As a consequence, it is required that the user provides an a priori information measure, which can then be refined as the robots gather more information about their surroundings. While applicable to some use cases, this is often a limitation of the existing ergodic controllers. It is not always the case that the information measure can be easily refined from the gathered data, or that an a priori information measure can be provided at all. Furthermore, it is also not always feasible to define an information measure for large-

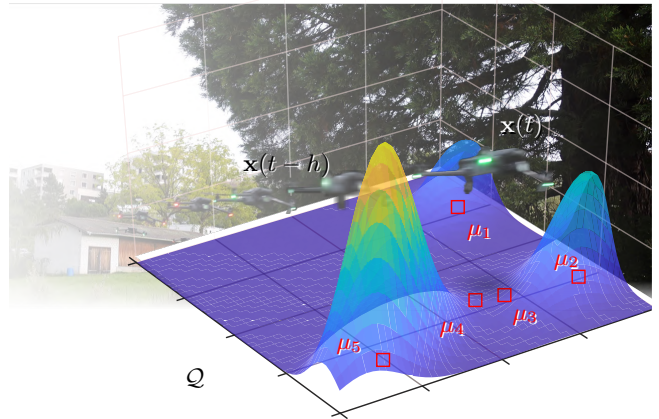


Fig. 1: Illustrative use case of our model agnostic ergodic controller. A drone explores an unknown state space using an Adaptive Gaussian Mixture Model, with the information density (3D plot) derived from the exploration history (points $\mathbf{x}(t-h)$ to $\mathbf{x}(t)$), dynamically avoiding obstacles (the tree).

scale and/or unknown spaces in practice, since constructing such a measure requires prior knowledge of the environment that may not be available. Here, with “large-scale” we refer to spaces on the order of dozens or even hundreds of meters.

Existing ergodic controllers have been studied from the point of view of time [7] and energy [8, 9] optimality and applied to a multitude of use cases. These use cases include tactile sensing [5], active learning [10], learning from demonstration [11], human-robot interaction [12, 13], multi-objective optimality [14, 15], grasping and manipulation [16, 17], and visual rendering [18, 19]. Ergodic controllers in the literature feature diverse aspects such as stochastic dynamics [20, 21] and multi-agent and/or swarm control with both centralized [8, 22] and distributed information processing [23, 24]. Although some of these use cases involve information gathering [25] and feature urban environments and other environments with potentially unknown underlying distributions [23, 26], a generic model agnostic ergodic controller has not been studied yet. Even though recent methods have been making progress in this direction [7–9, 27, 28], these methods do not scale efficiently due to the formulation of the underlying optimization (i.e., extending the state space requires a larger optimization horizon) and still require an a priori information measure. From an obstacle avoidance perspective, even though there are ergodic controllers that feature obstacle avoidance [23, 29], this is an external component on top of the ergodic controller that then results in a sub-ergodic solution [7] rather than an integral component of the explorer itself.

With this work, we address this challenge by introducing a

general and model-agnostic ergodic control methodology that is applicable beyond the standard formulations considered in prior work. In contrast to existing ergodic control methodologies that solely rely on external obstacle avoidance techniques, e.g., control barrier functions [29], RT-RRT* [23], etc., for obstacle avoidance capabilities, our methodology provides additionally a tunable degree of obstacle avoidance as a “by-product,” reducing the reliance on external obstacle avoidance techniques. The underlying information measure is represented utilizing an *Adaptive Gaussian Mixture Model (AGMM)*, which is derived from the sensory data as the robot traverses the state space – a process that is handled by our methodology and that does not require *any* user input. Our ergodic formulation is different from existing methods. The problem is posed so that the robot spends time in areas with low information measures, whereas the “explored space” is related to high information measures. The methodology is iterative and utilizes a model predictive controller (MPC) approach, where the ergodic controller is refined within a specified time window.

We showcase our general and model agnostic ergodic control on a large-scale problem with an unknown underlying information measure: the problem of exploring a simulated forest and an outdoor area that has both dense vegetation in an agricultural field (see Figure 1), in contrast to existing ergodic control methodologies that are demonstrated in structured environments or generally with a known information measure. Section IV shows the performance of our approach. The open-source software stack to replicate our approach and the experimental data are made available on our project repository webpage¹.

The remainder of this paper is structured as follows. Sec. II provides the details of the underlying principles and the problem addressed. Sec. III is split into two sub-sections: one provides an overview of ergodic control, whereas the other details our methodology. Sec. V provides conclusions and draws future perspectives.

II. PROBLEM FORMULATION

This work addresses the problem of exploring an unknown, bounded, and potentially large-scale state space. For practical reasons, we restrict the exploration space to one hectare. However, the formulation is such that the state space could potentially be unbounded.

Let us consider a bounded space $\mathcal{Q} \subset \mathbb{R}^2$. The robot is free to move in this space except for a finite number of obstacles represented by $\mathcal{O} \subset \mathcal{Q}$. The height of the robot is constrained by the relative height from the terrain. In the remainder, we utilize the concepts of ergodicity and ergodic metric to direct the robot into unexplored areas (i.e., with low information density) while avoiding the obstacles, i.e., $\mathcal{Q} \cap \mathcal{O}$, as opposed to other approaches featuring ergodic control in the literature where exploration happens in areas with high information density instead [4–6].

Definition II.1 (Ergodicity). Given the bounded state space $\mathcal{Q}$, a trajectory $\mathbf{x}(t) \in \mathcal{Q}$ is *ergodic* with respect to a spatial

distribution ϕ , or, analogously, is distributed among regions of high expected distribution, if and only if

$$\lim_{t \rightarrow \infty} \int_{\mathcal{Q}} \phi(\mathbf{x}) \Omega(\mathbf{x}) d\mathbf{x} = \frac{1}{t} \int_{\mathcal{T}} \Omega(\bar{\mathbf{x}}(t)) dt, \quad (1)$$

where $\bar{\cdot}$ is a map that maps the state space to the exploration spaces, and Ω are all the Lebesgue functions as defined in, e.g., [4].

The spatial distribution ϕ is derived using an Adaptive Gaussian Mixture Model.

Definition II.2 (Adaptive Gaussian Mixture Mode). Assume that there is a given number $n \in \mathbb{N}_{>0}$ of Gaussians $\mathcal{N}$ in a Gaussian Mixture Model, with an initial probability equally distributed. An *Adaptive Gaussian Mixture Mode* is

$$\phi(\boldsymbol{\alpha}, \boldsymbol{\mu}, \mathbf{x}) := \sum_{i=1}^n \alpha_i \mathcal{N}_i(\mathbf{x} | \mu_i, \Sigma_i), \quad (2)$$

where $\Sigma_i \in \mathbb{R}^{2 \times 2}$ indicates the covariance matrix and $\mu_i \in \mathcal{Q}$ denotes the center of a Gaussian $\mathcal{N}_i$.

The AGMM defined in this way has variable centers $\boldsymbol{\mu} \in \mathcal{Q}^n$ and variable mixing coefficients $\boldsymbol{\alpha} \in \mathbb{R}_{>0}^n$.

The notation $\mathbb{S}_{>0}$ denotes a strictly positive set $\mathbb{S}$. Bold letters indicate vectors, i.e., $\mathbf{x} \in \mathcal{Q}$ is the state space vector, whereas $\boldsymbol{\alpha}, \boldsymbol{\mu}$ are the vectors comprising the ideal AGMM’s mixing coefficients and position components respectively (see Sec. III).

An ergodic metric is then defined as a value that quantifies the ergodicity.

Definition II.3 (Ergodic metric). Consider a time average distribution of the trajectory over a limited time window t , e.g.,

$$h(\mathbf{x}(t)) := \frac{1}{t} \int_{\mathcal{T}} \Delta((\mathbf{x})(t)) dt, \quad (3)$$

where Δ is defined as a Dirac delta function. An *ergodic metric* is an L^2 -inner product in between the average of the spatial and time distributions.

Problem II.1 (Adaptive ergodic control). Given the state space and the obstacles space $\mathcal{Q}$ and $\mathcal{O}$ respectively, assume the number of Gaussian components n is given. The *adaptive ergodic control* problem is the problem of finding the evolution of the Gaussian components of an AGMM, i.e., $\boldsymbol{\alpha}(t)$ and $\boldsymbol{\mu}(t)$ and of the controls $\mathbf{u}(t) \in \mathcal{U}$ so that $\mathbf{x}(t)$ explores $\mathcal{Q}$ while avoiding $\mathcal{O}$ and the ergodic metric is minimized.

Note that it is not a requirement that the obstacle space $\mathcal{O}$ is known at the beginning of the exploration.

We propose a solution to Problem II.1 and demonstrate experimentally the feasibility of the solution, highlighting the trade-offs between accuracy (i.e., coverage) and exploration soundness (i.e., obstacle avoidance capabilities) in Sec. IV-A and IV-B respectively.

III. METHODS

This section details our methods. Sec. III-A introduces the concept of canonical ergodic control for exploration

¹Details omitted for double-blind reviewing.

in bounded areas with an information density distribution. Sec. III-B describes our methodology of adaptive ergodic control, i.e., ergodic control with an adaptive information density as a function of explored versus unexplored space.

A. Canonical ergodic control

To quantify the time average and the average of the spatial distributions ι and ϕ respectively, we use Fourier series basis functions, a common method for evaluating distributions in the spectral domain [4]. For the time average distribution, the coefficients of an equivalent basis function can be expressed as

$$c_k(\mathbf{x}(t)) := \int_{\mathcal{T}} \prod_{d \in \{1,2\}} \cos(2\pi k_d \mathbf{x}_d(\tau)/T) / T^2 d\tau / t, \quad (4)$$

where $T \in \mathbb{R}_{>0}$ is a given period and $\cdot_d$ denotes the d th item of a vector.

Equation (4) expresses the cosine basis function for a coefficient k , considering only the positive slice of the spectral domain and thus ignoring the function's imaginary component. The coefficients $k \in \mathcal{K}$ depend on a given number of frequencies $\kappa \in \mathbb{N}_{>0}$, including the base frequency, and are constructed so that $\mathcal{K} \in \mathbb{N}^2$ is a set of index vectors covering the set $\kappa \times \dots \times \kappa \in \mathbb{N}^{\kappa^2}$, i.e., they are built so that the coefficients are evaluated over the entire domain [30].

For the average of the spatial distribution, the coefficients of an equivalent basis function can be expressed similarly as

$$\phi_k(\mathbf{x}) := \int_{\mathcal{Q}} \sum_{d \in \{1,2\}} \phi(\mathbf{x}) c(\mathbf{x}) d\mathbf{x}, \quad (5)$$

where c is the integrand in Eq. (4) at the given point $\mathbf{x}$, evaluated at the current time step.

The aim of an ergodic controller is to minimize an ergodic metric, i.e., the L^2 -inner product of the distributions ι and ϕ (see Definition II.3). A consolidated metric [5, 7, 8, 10, 29, 31] for this purpose is, for instance,

$$\mathcal{E}(\mathbf{x}) := \sum_{k \in \mathcal{K}} \Lambda_k (c_k - \phi_k)^2 / 2, \quad (6)$$

where the coefficients of the time average and the average spatial distributions are expressed in Eq. (4-5).

Λ_k is a weight factor that determines the importance of different frequencies, e.g., with

$$\Lambda_k := \frac{1}{\sqrt{(1 + \|k\|^2)^3}}, \quad (7)$$

lower frequencies are preferred.

Note that in Eq. (5) we have utilized the expression for a standard Gaussian Mixture Model. We use the expression for the AGMM defined in Definition II.2 (i.e., Gaussian Mixture Model with variable centers and mixing coefficients) in the next section.

B. Adaptive ergodic control

To utilize the concept of an information density as a function of explored versus unexplored space, let us first consider Eq. (5) with the AGMM.

Assume for practical purposes that the space is square, with a given length $l \in \mathbb{R}_{>0}$ expressed in meters. We then restrict the period in Eq. (4) to this search space and express l as $T/2$. Eq. (5) can also be expressed as

$$\phi_k(\boldsymbol{\alpha}, \boldsymbol{\mu}, \mathbf{x}) := \int_{\mathcal{Q}} \left(\sum_{d \in \{1,2\}} \sum_{i=1}^n \alpha_i \mathcal{N}_i(\mathbf{x} | \bar{\boldsymbol{\mu}}_i, \bar{\boldsymbol{\Sigma}}_i) \right) c(\mathbf{x}) d\mathbf{x}, \quad (8)$$

where $\bar{\cdot}$ is a map that maps the center and the covariance matrix to a symmetric state space delimited by $-l$ and l , e.g., a map that uses linear transformation matrices [30].

We further define a value that represents the concept of ‘‘history.’’ If this value is expressed by $h \in \mathbb{R}_{>0}$, we can model the space that has already been explored by the robot using the definition of the AGMM. The covariance matrix can be expressed as

$$\Sigma_i := \frac{1}{2T} \int_{\mathcal{T}} \sum_{d \in \{1,2\}} (\mathbf{x}_d(\tau) - \mu_i) d\tau, \quad (9)$$

where the trajectory is evaluated within the history, i.e., Υ indicates the time interval between t and $t - h$.

The centers can then be expressed as $\mu_i := E(\mathbf{x}(t))$, with E being the expected value of $\mathbf{x}$ on Υ .

The adaptive ergodic control problem is thus framed as finding an ergodic controller that targets the inverse of the probability distribution represented by the AGMM, thereby avoiding areas ‘‘already visited’’ within a given history window h . However, this formulation may produce trajectories that require an additional obstacle avoidance methodology, such as [29].

Let us consider a modified expression for the center of the Gaussian

$$\mu_i := E(\mathbf{x}(t)) - e_i \quad (10)$$

where $e_i \in \mathbf{e} \subset \mathbb{R}^n$ represents a displacement that allows for ‘‘moving’’ the Gaussian components in the AGMM.

Our methodology ensures that the scaled ergodic controller determines the minimum displacement of the Gaussians so that the space to be visited is delimited by $\mathcal{Q} \cap \mathcal{O}$. This controller can be formulated as an optimal control problem (OCP)

$$\min_{\Theta} \mathcal{E}(\mathbf{x}) + \int_{\mathcal{T}} \mathbf{u}(\tau)^T R \mathbf{u}(\tau) d\tau + \Psi(\mathbf{e}), \quad (11a)$$

$$\text{s.t. } \dot{\mathbf{x}} = f(\mathbf{x}(t), \mathbf{u}(t)), \quad (11b)$$

$$\mathbf{x}(t) \in \mathcal{Q} \cap \mathcal{O}, \mathbf{u}(t) \in \mathcal{U}, \quad (11c)$$

$$\mathbf{x}(t_0), n, \kappa, l, h, \text{ are given}, \quad (11d)$$

where the output of the optimization Θ in Eq. (11a) is $\mathbf{x}, \mathbf{u}, \boldsymbol{\alpha}, \boldsymbol{\mu}$, i.e., the center of each Gaussian and its mixing coefficient in the AGMM, as functions of the control and the state. R is a control penalizing diagonal positive-definite

matrix. The function Ψ maps the displacement to a cost value, e.g.,

$$\Psi(\mathbf{e}) := \sum_{e_i \in \mathbf{e}} |e_i|, \quad (12)$$

where $|\cdot|$ is defined as an L^2 -norm.

The dynamics in Eq. (11b) represent a 2D single integrator system, which approximates the behavior of an unmanned aerial vehicle (UAV) in our experimental setup to a reasonable extent (see Sec. IV).

The problem is formulated to determine the displacement and probability for each Gaussian and the optimal (i.e., ergodic) control, ensuring that the displacement and probability deviate minimally from the ideal case. Note that the Gaussians represent the history of the explored space. For practical purposes, a defined horizon $N \in \mathbb{N}_{>0}$ is used, and the optimization is iterated for each horizon using a methodology similar to an MPC controller, where $\mathcal{T}$ in Eq. (4) denotes the interval between t and $t - N$ (with N not to be confused with the history window h).

Large-scale exploration is considered complete when a desired level of coverage is achieved (see Sec. IV). It is also possible to configure the problem so that exploration continues indefinitely or for extended periods, as discussed in [8].

Additional practical considerations, such as the choice of history and the size of the AGMM, are detailed in the next section.

IV. EXPERIMENTAL RESULTS

This section provides an overview of our experimental setup and showcases the results. Experiments are conducted using MATLAB (R), while interfaces for physical experimental evaluation are supported by a routine implemented in Python. The MPC optimization stack, which solves the OCP in Eq. (11), relies on two external open-source components: the non-linear programming solver IPOPT [32] and the algorithmic differentiation library CasADi [33].

In the following sections, Sec. IV-A presents experimental results from a simulated forest of 3600 square meters, serving as a representative scenario within the larger-scale problem discussed earlier. Sec. IV-B details our findings regarding the built-in obstacle avoidance capabilities of our general scaled ergodic controller compared to its coverage performance and discusses how the coverage evolves in time. Sec. IV-C presents our comparative experimental results utilizing another ergodic controller in the literature [30]. Sec. IV-D describes the field experiments, featuring an area with dense vegetation in an agricultural field, including multiple obstacles and a total area of 1 hectare.

A. Simulated forest

Simulated results are conducted in a forest with 45 trees randomly distributed across an area of 3600 square meters. The history h is set to 100 time steps (see Sec. IV-B) following a rigorous parameters identification where multiple sets of parameters have been evaluated for accuracy vs. obstacle avoidance capabilities (see Sec. IV-B). The simulation is

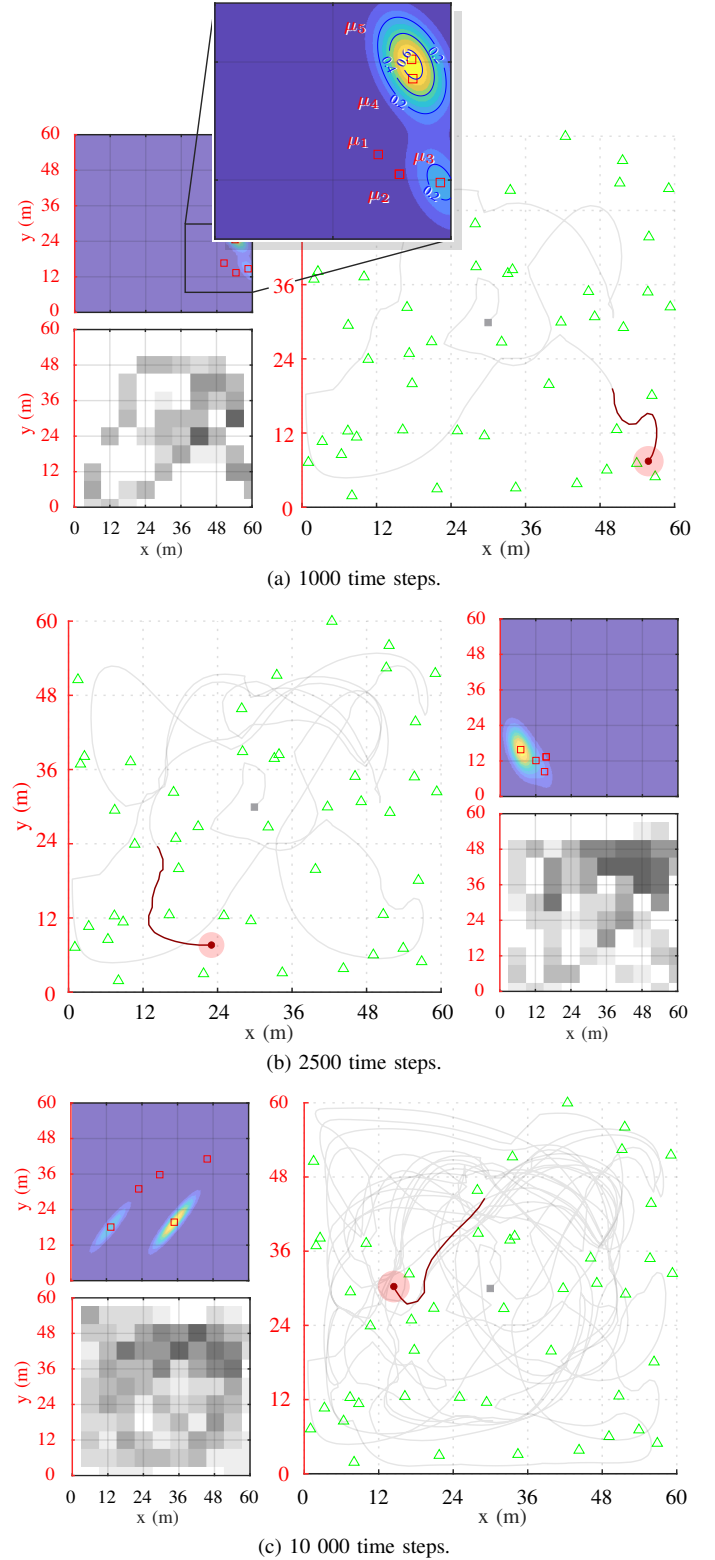


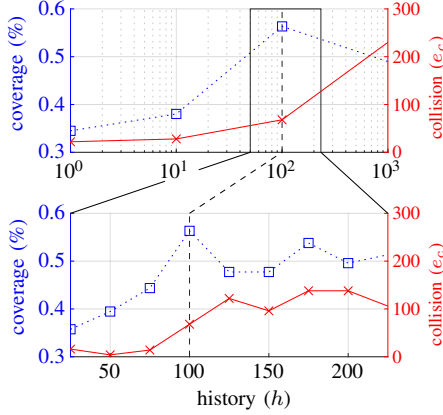
Fig. 2: **Exploration in a simulated forest with 45 trees over an area of 3600 square meters.** The figure shows the UAV's trajectory and the underlying information density distribution at various time steps. The coverage is reported with a heat map, i.e., darker areas have better coverage.

terminated at 50 000 time steps when a desired coverage threshold is achieved within a specified approximation (i.e., $60\% \pm 5\%$). The number of Gaussians n is set to five, with the initial positions centered in the middle of the state space.

Fig. 3: **Obstacle avoidance and coverage trade-offs.**

This figure illustrates the relationship between the size of the history window and two key metrics: obstacle avoidance and coverage. The collision metric e_c quantifies occurrences of constraint violations, with the right y-axis displaying the invalidation ratio for various history window sizes, from 25 to 225

(linear scale on the bottom) and from 1 to 1000 (logarithmic scale on the top). The coverage metric, calculated based on the area explored (excluding obstacles), is presented on the left y-axis. Optimal trade-offs are observed with a history window of approximately 100 (dashed black line), where the balance between obstacle avoidance and coverage is most effectively achieved.



The number of frequencies κ is set to nine, excluding the base frequency, consistent with other ergodic controllers in the literature [7, 8, 30].

Fig. 2 presents the results for the simulated forest at distinct time steps. Fig. 2a shows the trajectory (on the right) after 1000 time steps. The starting point is represented by the gray square. The history is illustrated with a dark red line, and the current point is marked by a dark red dot. The top-left of the figure depicts the underlying information density distribution (i.e., the AGMM from Definition II.2), including a detail of the five Gaussian components in the area representing the current history interval. The optimal values of the Gaussian centers μ and the mixing coefficients α ensure that the AGMM represents the history (i.e., the expected value and covariance of the trajectory over the current history window) while minimizing deviation to avoid obstacles. The bottom-left of the figure displays the coverage map, where each square counts the number of points within it at a given time.

Fig. 2b shows the trajectory at time instant 2500. The simulated UAV continues navigating through the obstacles, further exploring the state space. The AGMM is shown on the top right, and the coverage map is displayed at the bottom right.

Fig. 4: **Evolution of coverage over time for different history windows.** Larger history windows result in slower initial coverage due to the increased influence of past trajectory data but lead to more stable long-term coverage. Over time, the differences diminish as all configurations converge toward their respective steady-state behaviors.

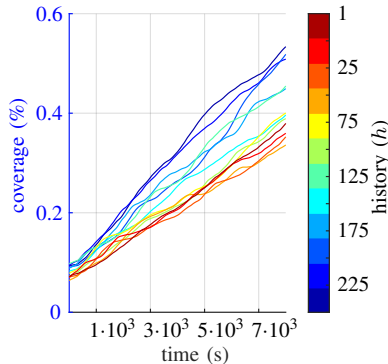


Fig. 2c shows the trajectory at time instant 10 000. As with the previous figures, the AGMM model and coverage map are illustrated at the top-left and bottom-left, respectively. It is observed that while the obstacles are generally avoided, the simulated UAV occasionally comes close to or passes through obstacles. A relationship between the degree of obstacle avoidance and coverage is reported, as discussed in the following section.

B. Obstacle avoidance vs. coverage

The OCP formulated in Eq. (11) may not always yield an optimal solution. In such cases, our algorithm provides a sub-optimal solution where the constraint in Eq. (11c) may not be fully respected. For example, in the simulated trees scenario described in Sec. IV-A, obstacles may not be avoided within the desired threshold. We observed this behavior during the initial iterations of our adaptive ergodic methodology. However, by tuning our approach using the concept of history h , we can improve obstacle avoidance as a by-product of our method.

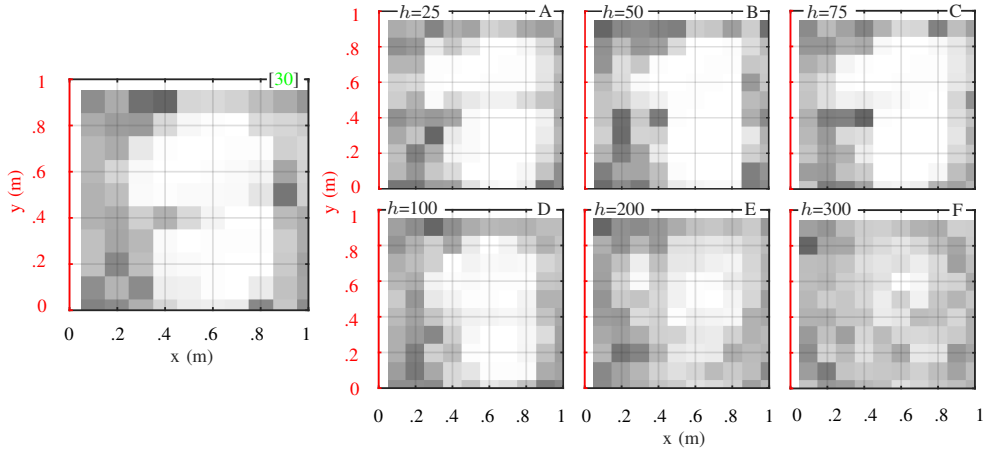
Fig. 3 presents our experimental results, highlighting the trade-offs between built-in obstacle avoidance capabilities and coverage, along with a collision metric. The collision metric e_c is designed to count every occurrence where a point violates the constraint in Eq. (11c) within a given threshold. The count of violations is shown on the left y-axis of the figure. With a horizon set to 50 000, the invalidation ratio for a history window of zero is only 0.044%, whereas it increases to 0.46% for a history window of 1000. The threshold for invalidation is set to 60 centimeters. Coverage is calculated as described in Sec. IV-A. Note that full coverage is not achieved since *obstacles are included in the overall coverage metric*.

The top of the figure shows the three metrics on a logarithmic scale. We observed that a history window of 100 provides the best trade-off between obstacle avoidance (0.14%) and coverage metrics. The bottom of the figure displays a detailed view of the history window around 100, ± 25 .

Fig. 4 shows the evolution over coverage in time for all the history windows in Fig. 3. Larger history windows tend to have a lower initial coverage as more trajectory history data have are processed to derive the adaptive ergodic controller, but they eventually lead to a more stable coverage rate as the exploration continues. The figure furthermore shows that the variance in coverage decreases over time, particularly for larger history windows.

The data demonstrate that our method inherently incorporates a tunable degree of obstacle avoidance. However, in practice, even low collision rates could lead to mission-critical failures. It is thus noted that our method still requires an external obstacle avoidance methodology in use cases where collisions lead to critical failures. Nonetheless, compared to other methods in the literature, our approach maintains optimality (ergodicity) across the majority of the state space, while *reducing* the reliance on external obstacle avoidance techniques.

Fig. 5: **Comparative results between the adaptive ergodic controller and the ergodic controller from [30].** The left-most figure replicates the coverage map from [30], where darker squares indicate areas of high information measure and lighter squares indicate areas of low information measure. The right-side figures illustrate the adaptive ergodic controller with different history windows A–F. The adaptive controller operates without an explicit underlying information measure, instead treating the original information distribution as two unknown obstacles that “are discovered upon proximity.”



C. Comparison with [30]

Comparative results are conducted against an existing ergodic controller in the literature to highlight the similarities between the derived information measure (ours) and the underlying information measure (literature). The underlying information measure (Fig. 5 in [30]) is modified so that the resulting ergodic trajectory remains in areas with low information measure, enabling a quantitative comparison.

The left-most figure in Fig. 5 illustrates the results as reported by [30] using a coverage map, where each square represents the number of points within it at a given time. Darker squares indicate areas where the controller spends more time, while lighter squares correspond to areas where the controller spends less time (see Sec. IV-A). The adaptive ergodic controller is then illustrated on the right, with different history windows A–F. Note that the adaptive ergodic controller is constructed without an underlying information measure. Instead, the original underlying distribution measure is treated as two unknown obstacles, each “discovered” when the controller comes into each obstacle’s proximity. History windows A–C illustrate the evolution of coverage with a history window of 50 ± 25 . The reported coverage is then 25.74%, 27.41%, and 30.89% for A, B, and C. D–F illustrate the evolution of coverage with a history window of 200 ± 100 . The reported coverage is then 20.39%, 20.74%, and 20.01% for D, E, and F. The history windows A–D all show relative high similarities with [30].

D. Outdoor model

Additional experimental evaluation is conducted using a model of an outdoor vegetated area located in Birmensdorf, ZH, Switzerland, with coordinates 47.362646° latitude and 8.457228° longitude. The area was selected to include dense vegetation in an agricultural field, simulating a real-world use case for our general adaptive ergodic controller. The exploration area is bounded to 1 hectare, and the surrounding region has been manually surveyed to assess future feasibility.

Fig. 6 illustrates the path taken, with the history represented by the dark red line and the current point marked by the dark red dot. The history is set to 150 time steps, and the dynamics in Eq. (11b) are adjusted to reasonably model a DJI (R) Mavic 3 UAV. The history is chosen to ensure

obstacle avoidance, allowing the UAV to navigate within the vegetation while avoiding tree trunks.

A preliminary physical evaluation was performed by manually setting the UAV along a subset of the planned waypoints, to assess the feasibility of the method for real-world use cases. The trajectory shown in Fig. 6, however, corresponds to the computed path from MATLAB (R). Further physical evaluation is ongoing (see Sec. V). Exploratory trajectories are imported into the UAV’s flight controller using waypoints managed by proprietary software; compatibility with other software and flight controllers is also supported. Data are reported after a horizon of 5000 time steps. The obstacles are positioned in the model so that the UAV operates within the canopy at an altitude of 2.5 meters, excluding small bushes but including tree trunks. The UAV’s velocity in the model is set to 1.5 meters per second.

V. CONCLUSION AND FUTURE DIRECTIONS

This paper addresses the problem of robotic exploration using ergodic control, overcoming limitations of existing ergodic controllers in terms of required information measure and scalability. We formulate the problem as one of adaptive ergodic control, where an AGMM represents the history of explored areas. Our methodology leverages an MPC-based ergodic controller to focus exploration on low-information density

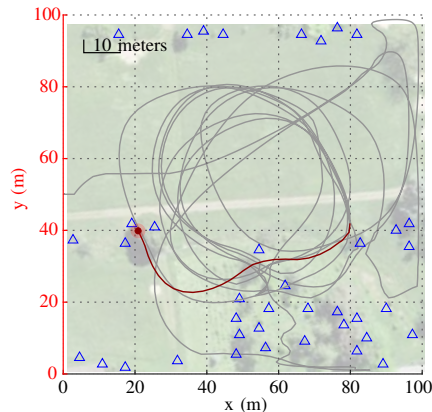


Fig. 6: **Outdoor model of a large-scale vegetated area featuring our ergodic controller.** The figure illustrates the UAV’s trajectory over 10 000 square-meter area located in Birmensdorf, Switzerland. The area features dense vegetation in an agricultural field. The dark red line represents the exploration history, while the dark red dot marks the UAV’s current position. The problem

is posed so that the modeled UAV operates at an altitude of 2.5 meters, avoiding tree trunks but not smaller bushes. The dynamics are adjusted to reflect real-world conditions, with the exploration history set to 150 time steps.

regions (as opposed to high-density areas) while minimizing deviations of the AGMM components and providing a tunable degree of obstacle avoidance as a “by-product.” A notable difference from previous work in ergodic control is that we *do not require an a priori user-defined information measure*, making our approach suited for exploration in unknown outdoor environments.

Our experimental results are conducted in various simulated and real-world environments, including a forest and dense vegetation in an agricultural field. The experiments highlight a balance between coverage efficiency and obstacle avoidance, with trade-offs being determined by the size of the history window, which is a given parameter that can be set depending on the desired value of coverage/obstacle avoidance. Comparative evaluation has been conducted against an existing ergodic controller in the literature and show that similar results are achieved. However, the existing ergodic controller requires an underlying information measure, whereas ours does not. Physical evaluation has been performed on an initial portion of the trajectory, and further physical experiments are ongoing: we are extending the results to a real-world agricultural use case, where ergodic controllers assist in detecting target species (i.e., pests and pathogens). Future work will focus on refining the physical experiments and integrating with a more varied range of flight controllers, aiming to improve long-term exploration capabilities. This includes employing energy-aware ergodic controllers and addressing planning and scheduling trade-offs for more specialized coverage use cases.

REFERENCES

- [1] M. Popović, T. Vidal-Calleja, G. Hitz, J. J. Chung *et al.*, “An informative path planning framework for UAV-based terrain monitoring,” *Autonomous Robots*, vol. 44, no. 6, pp. 889–911, 2020. [1](#)
- [2] L. Schmid, M. Pantic, R. Khanna, L. Ott *et al.*, “An efficient sampling-based method for online informative path planning in unknown environments,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 1500–1507, 2020. [1](#)
- [3] L. M. Miller, Y. Silverman, M. A. MacIver, and T. D. Murphey, “Ergodic exploration of distributed information,” *IEEE Transactions on Robotics*, vol. 32, no. 1, pp. 36–52, 2016. [1](#)
- [4] G. Mathew and I. Mezić, “Metrics for ergodicity and design of ergodic dynamics for multi-agent systems,” *Physica D: Nonlinear Phenomena*, vol. 240, no. 4, pp. 432–442, 2011. [1](#), [2](#), [3](#)
- [5] I. Abraham, A. Prabhakar, M. J. Z. Hartmann, and T. D. Murphey, “Ergodic exploration using binary sensing for nonparametric shape estimation,” *IEEE Robotics and Automation Letters*, vol. 2, no. 2, pp. 827–834, 2017. [1](#), [2](#), [3](#)
- [6] L. M. Miller and T. D. Murphey, “Trajectory optimization for continuous ergodic exploration,” in *IEEE American Control Conference (ACC)*, 2013, pp. 4196–4201. [1](#), [2](#)
- [7] D. Dong, H. Berger, and I. Abraham, “Time optimal ergodic search,” in *Conference on Robotics: Science and Systems (RSS)*, 2023, p. 13. [1](#), [3](#), [5](#)
- [8] A. Seewald, C. J. Lerch, M. Chancán, A. M. Dollar *et al.*, “Energy-aware ergodic search: Continuous exploration for multi-agent systems with battery constraints,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 7048–7054. [1](#), [3](#), [4](#), [5](#)
- [9] K. B. Naveed, D. Agrawal, C. Vermillion, and D. Panagou, “Eclares: Energy-aware clarity-driven ergodic search,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 14326–14332. [1](#)
- [10] I. Abraham, A. Prabhakar, and T. D. Murphey, “An ergodic measure for active learning from equilibrium,” *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 3, pp. 917–931, 2021. [1](#), [3](#)
- [11] A. Kalinowska, A. Prabhakar, K. Fitzsimons, and T. Murphey, “Ergodic imitation: Learning from what to do and what not to do,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 3648–3654. [1](#)
- [12] K. Fitzsimons, A. M. Acosta, J. P. A. Dewald, and T. D. Murphey, “Ergodicity reveals assistance and learning from physical human-robot interaction,” *Science Robotics*, vol. 4, no. 29, p. eaav6079, 2019. [1](#)
- [13] K. Fitzsimons and T. D. Murphey, “Ergodic shared control: Closing the loop on pHRI based on information encoded in motion,” *ACM Transactions on Human-Robot Interaction*, vol. 11, no. 4, 2022. [1](#)
- [14] Z. Ren, A. K. Srinivasan, B. Vundurthy, I. Abraham *et al.*, “A pareto-optimal local optimization framework for multiobjective ergodic search,” *IEEE Transactions on Robotics*, pp. 1–12, 2023. [1](#)
- [15] A. K. Srinivasan, G. Gutow, Z. Ren, I. Abraham *et al.*, “Multi-agent multi-objective ergodic search using branch and bound,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 844–849. [1](#)
- [16] S. Shetty, J. Silvério, and S. Calinon, “Ergodic exploration using tensor train: Applications in insertion tasks,” *IEEE Transactions on Robotics*, vol. 38, no. 2, pp. 906–921, 2022. [1](#)
- [17] C. Bilaloglu, T. Löw, and S. Calinon, “Whole-body ergodic exploration with a manipulator using diffusion,” *IEEE Robotics and Automation Letters*, vol. 8, no. 12, pp. 8581–8587, 2023. [1](#)
- [18] T. Löw, J. Maceiras, and S. Calinon, “drazBot: Using ergodic control to draw portraits,” *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 11728–11734, 2022. [1](#)
- [19] A. Prabhakar, A. Mavrommati, J. Schultz, and T. D. Murphey, “Autonomous visual rendering using physical motion,” in *Workshop on the Algorithmic Foundations of Robotics (WAFR)*. Springer, 2020, pp. 80–95. [1](#)
- [20] G. De La Torre, K. Flaßkamp, A. Prabhakar, and T. D. Murphey, “Ergodic exploration with stochastic sensor dynamics,” in *IEEE American Control Conference (ACC)*, 2016, pp. 2971–2976. [1](#)
- [21] E. Ayvali, H. Salman, and H. Choset, “Ergodic coverage in constrained environments using stochastic trajectory optimization,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 5204–5210. [1](#)
- [22] A. Rao, G. Sartoretti, and H. Choset, “Learning heterogeneous multi-agent allocations for ergodic search,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 12345–12352. [1](#)
- [23] A. Prabhakar, I. Abraham, A. Taylor, M. Schlafly *et al.*, “Ergodic specifications for flexible swarm control: From user commands to persistent adaptation,” in *Conference on Robotics: Science and Systems (RSS)*, 2020, p. 9. [1](#), [2](#)
- [24] H. Coffin, I. Abraham, G. Sartoretti, T. Dillstrom *et al.*, “Multi-agent dynamic ergodic search with low-information sensors,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2022, pp. 11480–11486. [1](#)
- [25] L. Dressel and M. J. Kochenderfer, “On the optimality of ergodic trajectories for information gathering tasks,” in *IEEE American Control Conference (ACC)*, 2018, pp. 1855–1861. [1](#)
- [26] A. Rao, A. Breitfeld, A. Candela, B. Jensen *et al.*, “Multi-objective ergodic search for dynamic information maps,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 10197–10204. [1](#)
- [27] E. Wittemyer and I. Abraham, “Bi-level image-guided ergodic exploration with applications to planetary rovers,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 10742–10748. [1](#)
- [28] S. Ivić, A. Sikirica, and B. Crnković, “Constrained multi-agent ergodic area surveying control based on finite element approximation of the potential field,” *Engineering Applications of Artificial Intelligence*, vol. 116, p. 15, 2022. [1](#)
- [29] C. Lerch, D. Dong, and I. Abraham, “Safety-critical ergodic exploration in cluttered environments via control barrier functions,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 10205–10211. [1](#), [2](#), [3](#)
- [30] S. Calinon, *Mixture models for the analysis, edition, and synthesis of continuous time series*. Springer, 2020, pp. 39–57. [3](#), [4](#), [5](#), [6](#)
- [31] I. Abraham and T. D. Murphey, “Decentralized ergodic control: Distribution-driven sensing and exploration for multiagent systems,” *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 2987–2994, 2018. [3](#)
- [32] A. Wächter and L. T. Biegler, “On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear program-

- ming,” *Mathematical Programming*, vol. 106, no. 1, pp. 25–57, 2006.
- ⁴
- [33] J. Andersson, J. Åkesson, and M. Diehl, “CasADi: A symbolic package for automatic differentiation and optimal control,” in *Conference on Recent Advances in Algorithmic Differentiation (AD)*. Springer, 2012, pp. 297–307. ⁴